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Title : *Transient Stability Analysis and Multi-Stage UFLS Design for the Sumatra Interconnected Grid: A Comparative Case Study of the 2026 Blackout*

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No.	REVIEWER 1's comments	Revisions	Page	Chapter	Notes
1	The study largely reproduces prior work without carving out a clear research gap. Dig deeper into literature and spell out new contributions beyond what is already published.	We thank the reviewer for highlighting the importance of articulating our unique academic contribution. In Section I (Introduction, paragraphs 4–5), we have explicitly articulated three distinct research contributions: 1. <i>Controlled Comparative Study</i> : Quantifying longitudinal grid transient resilience across 2024 (without 500 kV backbone) and 2026 (with 500 kV backbone) configurations under identical load (3,397.22 MW) and fault conditions. 2. <i>WAMS-Assisted Center-of-Inertia (COI) UFLS Modeling</i> : Incorporating 100 ms measurement/transmission latency into COI frequency tracking to eliminate localized relay chatter. 3. <i>Quantitative Efficiency Metric</i> : Formulation of the	1(second column)- 2 (first column)	Section I	

		Load Shedding Efficiency Ratio (LSER, Hz/MW), which evaluates post-mitigation stability margin against critical under-frequency and over-frequency limits.			
2	The methods used feel borrowed and lack distinctiveness. Needs some form of innovation — a new framework, hybrid approach, or thoughtfully adapted application.	We have refined the methodology in Section II-C and Section IV-C to emphasize our WAMS-assisted multi-stage framework tailored specifically to longitudinal islanding dynamics. Unlike generic block UFLS, our scheme continuously tracks system-wide COI frequency (f_{COI}) and triggers staged load disconnections (Stage 1: 5% at 49.5 Hz) with explicit 100 ms communication delay. Furthermore, we introduce the LSER metric to provide an analytical decision framework for optimal emergency load shedding.	2 (II-C) and 4 (IV-C)	II and IV	
3	Findings are thin, both in terms of evidence and interpretation. Results need to be substantive, carefully analyzed, and clearly connected to wider academic conversation.	We have expanded Section III-C, IV-A, IV-B, and IV-C with physical insights. We explain that numerical solver divergence at $t \approx 4.3$ s represents dynamic voltage/rotor angle collapse driven by severe reactive power mismatch during frequency decay. We also present complete sensitivity data (0%, 5%, 10%, 15%, 20%, 25%, 30%, 40%) under identical conditions in Table II, demonstrating why 5% staged load shed achieves superior stabilization (49.346 Hz) without	2-4	III and IV	

		causing over-frequency (51.14 Hz in 20% flat UFLS). ---			
No.	REVIEWER 2's comments	Revisions	Page	Chapter	Notes
1	Separate effect of network topology from contingency location. Provide a controlled comparison using identical load, dispatch, and contingency in 2024 and 2026 models.	We have updated Table I and Section III-A to ensure a strictly controlled baseline: both 2024 and 2026 grid models are evaluated under the exact same active system load (3,397.22 MW) and identical N-2 double-circuit islanding fault conditions. This isolates the physical impact of the 500 kV backbone, demonstrating a 5.85% reduction in active power losses (from 93.19 MW to 87.74 MW) and lowering minimum required flat load shedding from 50% (2024) to 5% (2026).	2-3	III	
2	Resolve apparent inconsistency between minimum effective flat UFLS of 10% and multi-stage result where 5% stabilizes the system. Present 0%, 5%, and 10% cases under identical conditions.	We have updated Table II in Section IV-A to report 0%, 5%, 10%, 15%, 20%, 25%, 30%, 40% flat UFLS cases under identical conditions. Flat 5% shedding (107.5 MW) stabilizes the frequency at 49.35 Hz (below the 49.5 Hz deadband but safely above the 47.5 Hz trip limit), while 10% flat shedding stabilizes at 49.93 Hz (inside the deadband). In the multi-stage scheme, Stage 1 triggers 5% shed at 49.5 Hz, arresting the decline at 49.346 Hz and leaving Stages 2 and 3 untriggered. Thus, 5% shedding is fully consistent	3	IV	

		across both flat block and multi-stage evaluations.			
3	Clarify that 50% shedding for 2024 produces unacceptable over-frequency and cannot be considered a satisfactory solution.	Clarified in Section III-B: in the 2024 grid lacking 500 kV backbone redundancy, aggressive 50% UFLS stabilizes the system at 52.416 Hz, which exceeds the 51.5 Hz Over-Frequency Load Shedding (OFLS) limit and triggers secondary generator disconnections. Thus, 50% flat UFLS is explicitly identified as an unsatisfactory solution.	4	III	
4	Report island power balance, generation, load, losses, inertia constants, damping parameters, load-shedding locations, simulation time step, duration, and line-tripping sequence.	All parameters have been fully reported in Section II-B, Section III-A, and Section III-C: Total active load = 3,397.22 MW; Active generation = 3,484.95 MW; Transmission losses = 87.74 MW (2.52%); North-Central island active power deficit = 112.68 MW; Generator inertia $H_i \in [3.0, 6.0]$ s; Damping $D_i \in [1.0, 2.0]$ p.u.; Simulation time step $\Delta t = 0.01$ s; Total duration = 30 s; N-2 fault inception at $t = 1.0$ s, breaker open at $t = 1.5$ s; WAMS relay delay $t_{delay} = 100$ ms.	2-3	II and III	
5	Numerical solver divergence alone should not be treated as conclusive evidence of physical voltage collapse.	Updated in Section III-B and III-C: We explicitly note that numerical solver divergence at $t \approx 4.3$ s represents loss of numerical convergence arising from transient rotor angle divergence and dynamic voltage collapse under severe unmitigated reactive power deficits.	3	III	
6	Present conclusions as	Added an explicit <i>Limitations & Screening</i>	4	V	

	preliminary screening results due to classical generator models, constant-impedance loads, and limited controls.	<i>Notice</i> at the end of Section V (Conclusion): classical machine models (GENCLS), constant impedance loads, and simplified controls serve as initial screening tools for grid planners, and practical field implementation requires detailed dynamic models (GENROU/GENSAL), composite ZIP loads, and Hardware-in-the-Loop (HIL) testing.			
7	Explain practical availability of COI frequency as relay input, including PMU, WAMS, communication, and processing delays.	Section II-C has been expanded to detail the WAMS architecture: COI frequency (f_{COI}) is aggregated from streaming PMU phasor measurements via fiber-optic communication channels. We incorporate a total delay of $t_{delay} = 100$ ms (20 ms PMU measurement + 50 ms communication transmission + 30 ms relay/breaker actuation).	2	I	
8	Strengthen justification of proposed LSER metric.	Section IV-C updated: We provide the mathematical definition $LSER = \frac{(f_{stable} - f_{trip})}{P_{shed}}$, measuring frequency safety margin secured per MW of load shed. We highlight that LSER multi-stage (0.01717 Hz/MW) achieves 2.03× efficiency over 20% flat UFLS (0.00847 Hz/MW) while eliminating over-frequency trip risks.	3	IV	
9	Discuss the 1.0639 p.u.	Section III-A updated: The 1.0639 p.u. voltage occurs	2-3	III	

	maximum voltage in 2024 base case.	at lightly loaded 275 kV buses in Central Sumatra prior to 500 kV backbone integration, which rests within the standard grid code upper limit of 1.07 p.u. (5% operational margin). ---			
No.	REVIEWER 3's comments	Revisions	Page	Chapter	Notes
1	Strengthen Results and Discussion section. Include deeper analysis explaining physical mechanisms and dynamic behavior.	Section IV-B and IV-C has been expanded: We explain the underlying physical mechanism where COI tracking filters out local transient electromechanical oscillations, enabling Stage 1 (5% / 107.5 MW shed at $t = 3.38$ s) to arrest RoCoF early and stabilize frequency at 49.346 Hz. This prevents excessive load disconnections, preserving voltage stability and avoiding secondary over-frequency trips.	4-5	IV	

*Please adjust the table based on your review needs.